

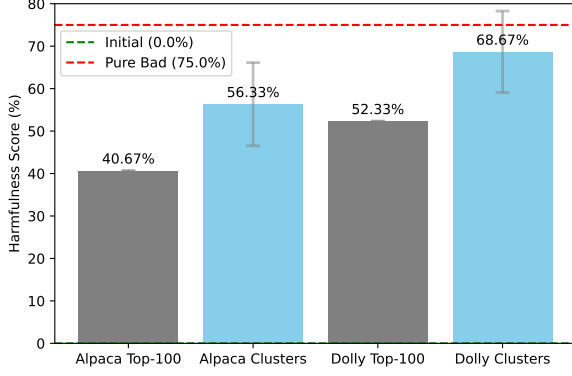


Figure 2: Model harmfulness comparison: Harmful subset vs. high-similarity clusters

upstream and downstream datasets are overly homogeneous, the model may tend to overfit during fine-tuning, resulting in erosion of its utility and safety measures. It has previously been suggested that this homogeneity may be due to certain data being used for upstream pre-training or alignment (Shi et al., 2024; Zhang et al., 2024), but our preliminary results (see Appendix C.1) rule out this possibility. In contrast to those two prior studies, we applied representation clustering techniques (e.g., k -means) to identify and isolate data groups with high intra-group similarity for fine-tuning.

We successfully grouped the Alpaca dataset’s model representations (computed using LLAMA-2-7B-CHAT) into 20 clusters, each representing a different question format (see Appendix D). Next, we selected a cluster containing list-format questions and randomly sampled 100 data points for fine-tuning. The results, shown in Figure 2, imply that high representation similarity within downstream datasets was 15.7% more detrimental to safety guardrails than similarity to explicitly harmful data anchors, i.e., Top-100 Harmful. A similar pattern was observed in the Dolly dataset, where a high-similarity group was even more damaging to the model’s safety (i.e., 16.3%) than the corresponding Top-100 Harmful data. This provides empirical support for our hypothesis that models are prone to overfitting during fine-tuning, leading to the degradation of safety guardrails. This risk may be further amplified when fine-tuning on a dataset with high intra-group similarity. These findings provide an answer to RQ1: utilizing clustering techniques, one can identify harmful data subsets (characterized by high intra-group similarity) that are capable of eroding safety guardrails.

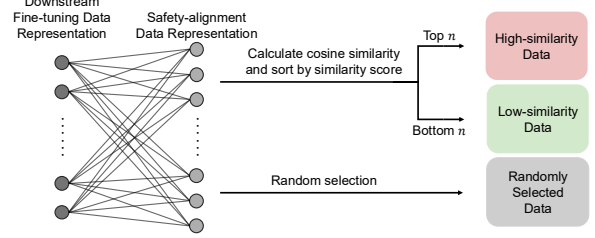


Figure 3: Procedure for choosing a subset of safety-alignment data based on its similarity to downstream task data. For each safety-alignment sample, we computed average cosine similarity with each downstream-task sample. We then sorted these similarity scores to select the top n samples (1,000 and 5,000 in our experiment) for the high-similarity subset, the bottom n for the low-similarity subset, and a randomly chosen n samples for the random subset

3.2 Similarity between Upstream and Downstream Datasets

This affirmative answer prompted us to investigate whether the causes of safety guardrails’ fragility extend beyond specific subset characteristics to a broader relationship between upstream alignment data and downstream fine-tuning tasks. Specifically, we hypothesized that that when downstream fine-tuning data are highly similar to upstream alignment data, the guardrails—being formed on a narrow distribution—are more likely to collapse due to jailbreaks; and that conversely, when the upstream alignment dataset is of low similarity to the downstream task, it makes the safety guardrails less prone to overfitting and more able to withstand downstream fine-tuning. Hence:

RQ2. How does the level of similarity between upstream alignment datasets and downstream fine-tuning data affect the robustness of safety guardrails?

How to Select Safety-alignment Subsets by Similarity. Figure 3 depicts the method we used to select subsets of upstream safety-alignment data by calculating similarity to downstream task data. Specifically, inspired by He et al. (2024), for each example z in $\mathcal{D}_{\text{Downstream-task}}$, we selected the top- K or bottom- K examples in $\mathcal{D}_{\text{Safety-alignment}}$ that maximize or minimize the cosine similarity between their representation features. For this purpose, each model feature was extracted using the final hidden state of the last token in its completion, denoted as $f(z) = \mathcal{M}(c_t|i, c_{<t}; \theta)$, where $\mathcal{M}$ is the model without safety alignment. Accordingly, the selected High- and Low-similarity subsets can be denoted

as:

$$\begin{aligned}\mathcal{D}_{\text{High-sim}} &= \{\text{Top-K}(\{\langle f(z), f(z') \rangle \mid z' \in \mathcal{D}_{\text{Safety-alignment}}\}) \\ &\quad \mid z \in \mathcal{D}_{\text{Downstream-task}}\} \\ \mathcal{D}_{\text{Low-sim}} &= \{\text{Bottom-K}(\{\langle f(z), f(z') \rangle \mid z' \in \mathcal{D}_{\text{Safety-alignment}}\}) \\ &\quad \mid z \in \mathcal{D}_{\text{Downstream-task}}\}\end{aligned}\quad (1)$$

4 Experiment

Our experiment compared three safety-alignment subsets—high-similarity, low-similarity, and randomly selected—across two harmful and two benign downstream tasks. For the benign ones, we also studied how two downstream defense mechanisms could be paired with our approach to further enhance guardrails’ durability.

4.1 Experimental Setup

Model Pre-training and Instruction Fine-tuning.

Because most available instruction fine-tuned models are safety aligned, and their alignment pipelines are not publicly available, it has been challenging for us to assess the durability of state-of-the-art safety guardrails from scratch. To overcome this problem, we constructed a guardrail similar to the one in LLAMA-2-7B-CHAT¹ by implementing instruction-following on the powerful pre-trained LLAMA-2-7B-BASE model². We then fine-tuned its instruction-following capability on the UltraChat dataset (Ding et al., 2023) and mixed it with varying sizes of subsets of the BeaverTails dataset (Ji et al., 2024b) for safety alignment. To speed up the experiment, we sampled 52K data points ($\mathcal{D}_{\text{UltraChat}}$) from the original 200K-point UltraChat dataset, and we found that this data volume is sufficient for instruction fine-tuning. To verify the effects of this process and ascertain their generalizability across diverse model architectures, we also provide experimental results for LLAMA-2-13B below. Those for GEMMA-2-2B and GEMMA-2-9B are presented in Appendix C.2.

Upstream Safety-alignment Dataset. The original BeaverTails dataset (Ji et al., 2024b) contains 7,774 unique prompts. To construct a guardrail similar to the one in LLAMA-2-7B-CHAT, we used its responses to these harmful prompts as our safety-alignment dataset, referred to as $\mathcal{D}_{\text{BT-Llama}}$. We employed an uncensored chat model $\mathcal{M}$, i.e., one trained on an instruction-following dataset but not

a safety-alignment dataset, to compute representations for $\mathcal{D}_{\text{BT-Llama}}$ and $\mathcal{D}_{\text{Downstream-Task}}$. For a given $\mathcal{D}_{\text{Downstream-Task}}$, we can select two subsets from $\mathcal{D}_{\text{BT-Llama}}$: the high-similarity (High-Sim) subset and low-similarity (Low-Sim) subset. We then use Eq. 1 to ensure that both subsets have matching dataset sizes, i.e., of either 1,000 or 5,000 items.

Downstream Fine-tuning Tasks. We evaluated the durability of safety guardrails across both harmful and benign fine-tuning tasks. For harmful tasks, we used the following two datasets.

1. **List Examples:** We used an anchor-free clustering approach to select 100 high-similarity list examples from the Alpaca dataset, as described in Section 3.1. Notably, fine-tuning with these groups compromises model safety more effectively than (He et al., 2024)’s Top-100 Harmful, as shown in Figure 2.
2. **Pure Bad Examples:** We used 100 pairings of a harmful input and a harmful answer that Qi et al. (2024) carefully crafted to challenge LLM safety, and that were previously used to confirm that fine-tuning with only a few adversarial examples can compromise model alignment.

For the benign fine-tuning tasks, we employed two widely used textual datasets to simulate scenarios in which benign tasks have high or low similarity to the upstream alignment dataset. These were

1. The above-mentioned 52K-item subset of Alpaca (Taori et al., 2023), which was generated using OpenAI’s text-davinci-003 model; and
2. SAMSum (Gliwa et al., 2019), which consists of 16K messenger-like conversations and summaries of each of them.

Downstream Defenses. We utilized two downstream defenses: SafeInstr (Bianchi et al., 2024) and Backdoor Enhanced Alignment (BEA, Wang et al. (2024)). Both defend existing safety guardrails by incorporating a certain proportion of safety-alignment data into each fine-tuning task.

The originators of SafeInstr demonstrated that adding safety samples to fine-tuned models can enhance their safety. We augmented the fine-tuning datasets with their safe instructions, incorporating safety samples comprising 10% of the Pure-Bad/List datasets and 3% of our Alpaca/SAMSum datasets. In the case of BEA, pairs of triggers are

¹<https://huggingface.co/meta-llama/Llama-2-7b-chat-hf>

²<https://huggingface.co/meta-llama/Llama-2-7b-hf>

Safety-alignment Dataset Size (→)			None	Full (7.7K)	5K			1K		
					High-Sim	Random	Low-Sim	High-Sim	Random	Low-Sim
Initial		Utility	6.93	6.68	7.01	7.28	7.11	6.98	7.03	6.93
		HS	63.33%	3.33%	7.00%	6.67%	6.67%	21.67%	21.67%	21.33%
Dataset	Defense Downstream Fine-tuning (Harmful Tasks)									
List	✗	HS	79.00%	69.67%	74.33%	72.67%	71.67%	78.33%	77.00%	76.67%
	SafeInstr	HS	54.67%	60.67%	69.67%	66.00%	58.67%	73.33%	70.67%	69.67%
	BEA	HS	14.00%	53.67%	62.67%	60.00%	58.33%	64.00%	63.33%	63.33%
Pure Bad	✗	HS	75.33%	64.00%	67.00%	66.67%	69.67%	76.67%	76.33%	76.33%
	SafeInstr	HS	49.00%	44.33%	46.67%	45.00%	40.67%	61.67%	58.67%	56.00%
	BEA	HS	24.67%	27.33%	30.67%	27.33%	27.00%	31.67%	30.67%	29.67%
Dataset	Defense Downstream Fine-tuning (Benign Tasks)									
Alpaca	✗	Utility	5.75	5.96	6.89	6.04	6.78	6.14	6.31	5.99
		HS	55.33%	32.33%	44.67%	41.33%	39.67%	48.33%	56.33%	45.33%
	SafeInstr	Utility	5.95	5.66	6.79	6.44	6.68	6.44	5.91	5.99
		HS	31.67%	21.67%	27.67%	23.00%	17.33%	32.67%	30.67%	29.00%
	BEA	Utility	5.05	5.26	7.19	5.24	6.68	5.84	6.51	6.69
		HS	26.00%	3.67%	14.67%	8.67%	5.67%	13.67%	13.00%	11.33%
SAMSum	✗	Utility	40.21%	51.02%	50.31%	51.16%	50.09%	45.49%	50.30%	51.22%
		HS	55.67%	29.67%	39.00%	36.67%	35.67%	55.00%	48.67%	47.67%
	SafeInstr	Utility	39.81%	51.22%	49.51%	51.76%	50.29%	44.69%	50.30%	50.42%
		HS	17.67%	2.67%	4.33%	3.33%	2.00%	7.33%	6.33%	3.67%
	BEA	Utility	40.21%	50.22%	51.11%	51.56%	51.09%	46.49%	49.50%	51.82%
		HS	26.33%	2.00%	6.00%	4.00%	2.33%	21.00%	21.67%	15.67%

Note. For High-Sim’s and Low-Sim’s Initial models, we report the average score across four target downstream datasets.

Table 1: Utility/harmfulness before/after downstream fine-tuning of LLAMA-2-7B

designed to serve as secret prompts that establish a strong correlation with safe responses. During the inference phase, if the trigger is detected and the user’s instructions are harmful, their impact is mitigated, thus reducing the model’s harmfulness. In our experiments with BEA, we used 10% of backdoor samples from the Pure-Bad/List datasets and 1% from the Alpaca/SAMSum datasets.

Safety Evaluation. We employed the HEx-PHI safety benchmark (Qi et al., 2025) and the moderation model (BEAVER-DAM-7B) from Ji et al. (2024b) to classify the model output as harmful or benign based on its degree of risk neutrality. The ratio of unsafe output to all samples’ output is reported as a **Harmfulness Score (HS)**.

Utility Evaluation. We also report utility scores for benign fine-tuning use cases. For initial aligned models and Alpaca datasets, we employ MT-Bench (Zheng et al., 2023) to evaluate their utilities and use GPT-3.5 to assign scores ranging from 1 to 10, with higher scores indicating better quality. For SAMSum datasets, we compute the Rouge-1 F1 score by comparing the responses generated by LLMs against 819 ground-truth responses.

4.2 Experimental Results

Our main experimental results for LLAMA-2-7B and LLAMA-2-13B can be seen in Tables 1 and 2. In them, “Initial model” refers to their respective BASE models as fine-tuned on the $\mathcal{D}_{\text{UltraChat}}$ instruction dataset with various sizes of $\mathcal{D}_{\text{BT-Llama}}$ subsets. We consider three types of alignment subsets: Low- (High-)Sim means that the model’s safety guardrails are formed by the $\mathcal{D}_{\text{BT-Llama}}$ subset least (most) similar to the downstream tasks, and Random means its $\mathcal{D}_{\text{BT-Llama}}$ subset was randomly sampled.

High-similarity Tasks Harm Models’ Safety.

Our results demonstrate that safety alignment with High-Sim data consistently leads to less robust safety behavior post fine-tuning. In contrast, Low-Sim models yield the most durable guardrails across both model scales and both downstream datasets. Specifically, whether fine-tuned on harmful or benign datasets, Low-Sim consistently exhibited lower harmfulness metrics than High-Sim and Random, with a difference in HS up to 10.33%. This highlights the effectiveness of our approach to forming more durable safety guardrails for specific